

Brandon Oltarsh

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Education

Purdue University, Class of 2028

Major: Aerospace Engineering

GPA: 4.0

Portfolio

<https://brandonprog.github.io/brandonsportfolio/>

Experience

FRC Robotics – (January 2022-April 2024)

Mechanical Director Team 2383

- Leading workshops and design classes to teach underclassmen important concepts of mechanical design and controllability.
- Maintaining a large lab and machine shop hosting workdays of over 60 people.
- Designing and building large efficient robots for competition and outreach events in a team setting.
- Presenting in groups with peers for outreach awards.

Purdue Space Program: Active Controls

Cooling analysis/Engine optimization

- Using Matlab to analyze regenerative cooling of a 500-pound force liquid rocket engine
- Developing a method to analyze the effects of film cooling
- Comparing different materials and alcohol based fuels for efficiency and cooling properties

Robotics Summer Camp-(2022-2024, 4 weeks each summer)

Design Teacher/Senior Counselor

- Teaching children ages 10-14 basic design using Onshape with lectures, challenges, and group events.
- Facilitating pick-up and drop off of hundreds of children every day from multiple summer camps.
- Designing games for children to play and robot bases for them to build upon.

Craigs Pantry-(August 2018-May 2024)

Volunteer (Every other Sunday)

- Packaging and delivering food for food-insecure population of Broward County.
- Facilitating large donations to the charity from the local community including hundreds of pounds of food and hundreds of articles of clothing.

Skills

- 3D Modeling/CAM (Onshape, Autodesk Inventor)
- CNC Machining (100+ hours experience on a CAMaster Cobra)
- FDM 3D Printing (experience with many commercial slicers)
- Python, HTML, CSS
- Excel, Word, PowerPoint